

ORIGINAL RESEARCH

A parking occupancy prediction method incorporating time series decomposition and temporal pattern attention mechanism

Wei Ye¹  | Haoxuan Kuang¹ | Jun Li¹ | Xinjun Lai² | Haohao Qu¹

¹School of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen, People's Republic of China

²Guangdong CIM Provincial Key Lab, School of Electro-Mechanical Engineering Guangdong University of Technology, Guangzhou, People's Republic of China

Correspondence

Jun Li, School of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen 518107, People's Republic of China.

Email: stsljun@mail.sysu.edu.cn

Funding information

Guangdong Key Areas R&D Program, Grant/Award Number: 2019B090913001

Abstract

Parking occupancy prediction is an important reference for travel decisions and parking management. However, due to various related factors, such as commuting or traffic accidents, parking occupancy has complex change features that are difficult to model accurately, thus making it difficult for parking occupancy to be accurately predicted. Moreover, how to give appropriate weights to these changing features in prediction becomes a new challenge in the era of machine learning. To tackle these challenges, a parking occupancy prediction method called time series decomposition–long and short-term memory neural network (LSTM)–temporal pattern attention mechanism, which consists of three modules, namely 1) time series decomposition: modelling parking occupancy changes by extracting features such as trend, period, and effect; 2) encoder: extracting temporal correlations of feature sequences with LSTM; 3) temporal pattern attention mechanism: assigning attention to different features, are proposed. The evaluation results of 30 parking lots in Guangzhou city show that the proposed model 1) improves accuracy over the baseline model LSTM by 9.14% on average; 2) performs outstanding in four prediction time intervals and six types of parking lots, proving its validity and generality; 3) demonstrates its rationality and interpretability through ablation experiments and Shapley additive explanation.

1 | INTRODUCTION

With the massive growth in the number of vehicles, the limited parking spaces and the traffic congestion caused by parking problem have become one of the major problems of urban traffic. Researches show that cruising for parking spaces results in a peak increase of about 25–40% in the traffic flow [1], the average time spent finding a parking space is 3.5–14 min [2] and cruising for parking spaces wastes gasoline and produces more pollution emissions every year [3]. In other words, cruise for parking has become a common phenomenon in periods and areas with dense parking demand, which not only increases travellers' overall travel cost, but also adds additional congestion and emissions [4]. To tackle this issue, numerous parking management and guidance systems have been developed in the past decades. A reliable and accuracy real-time parking occupancy prediction (POP) is one of the key technologies of the parking management and guidance system, it not only can make proper

parking location recommendations and navigation in advance but also can help the parking authorities adjust the parking rate dynamically to improve the overall efficiency of the parking resources [5].

POP can be considered as a time-series prediction problem, which influenced by various environmental factors and travel behaviour. For example the cyclical pattern in rush hours caused by commuting parking behaviour in office and residential areas [6]. Moreover, the effect of events, such as extreme weather, traffic accidents, or just festivals, can cause dramatic changes in parking occupancy as well. To model these features of parking occupancy is an inevitable challenge for urban transportation researchers. Furthermore, how to give a set of appropriate weights to these features in prediction has also become an emerging challenge we must face in the age of machine learning.

In the current literature, there are two types of methods to extract the change features of parking occupancy to improve the prediction accuracy: 1) data augmentation by generating

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2023 The Authors. *IET Intelligent Transport Systems* published by John Wiley & Sons Ltd on behalf of The Institution of Engineering and Technology.

fake samples that look like original data, introducing extra heterogeneous data, or decomposing patterns [7–10]; 2) structure optimization by using advanced machine learning or deep learning methods [11, 12]. However, these methods still suffer from two major shortages. First, it is difficult to comprehensively extract multiple features of parking occupancy, such as cyclical patterns and the events effect of traffic accidents, holidays or weather. Second, in existing parking occupancy prediction methods, the influence degree of different features on the prediction results has not been considered.

To overcome these shortages, we proposed an approach named TSD-LSTM-TPA, which optimizes the LSTM network from two aspects, 1) data enhancement: we adopt time series decomposition (TSD) [13] to model trend feature, cyclical feature, and effect feature of parking occupancy; 2) structure optimization: we introduce time pattern attention mechanism (TPA) [14] to assign attention to different features extracted by TSD so that the method can make an accurate real-time prediction.

In particular, the main contributions of the study are as follows:

- We propose a parking occupancy prediction method based on the integration of TSD, LSTM, and TPA. Specifically, the adoption of the TSD module enables the LSTM network to extract the temporal features more efficiently. Correspondingly, the TPA module gives a set of reasonable weights (i.e. attention) to each temporal feature. Thus, our method can not only predict parking occupancy accurately and robustly but also be interpretable.
- We tested the accuracy, generalizability, and interpretability of the proposed method on a real-world shared dataset with 30 parking lots in Guangzhou, China. The results show that TSD-LSTM-TPA achieves outstanding performance in four prediction time intervals (i.e. 15, 30, 45, and 60 min) and six types of parking lots (i.e. commercial buildings, hospitals, office buildings, sports, recreational facilities, tourist sites, and residential areas). Moreover, TSD-LSTM-TPA improved accuracy over the baseline model LSTM by 9.14% on average.
- We found that the trend and cycle features play a major role in prediction and quantified the importance of features in each input time step through ablation experiment and Shapley additive explanations (SHAP) [15], respectively.

The rest of this paper is organized as follows: Section 2 summarizes the literature review; Section 3 introduces the proposed parking occupancy prediction method; Section 4 presents the experiments in 30 parking lots, including descriptions of the data set, results of the experiments, and findings; Finally, Section 5 concludes the paper and discusses possible future work.

2 | LITERATURE REVIEW

Parking occupancy prediction is a key technology for parking management and parking guidance systems, which is also

a typical time series prediction problem. In order to improve the method's performance for POP without introducing any other data sources, there are two popular solutions, i.e. data augmentation and structure optimization.

Data augmentation has always been playing an important role in parking occupancy prediction, whose key idea is to decompose original data into several high-dimension patterns through statistical methods, so that it is easier for feature learning [16, 17]. For example CoPASample [18], which generates samples that reflect the first- and second-order statistical information of the dataset, thus augmenting the dataset in a way that preserves the total covariance of the data. Another illustration is an enhanced tensor factorization model, which incorporates transportation system knowledge and uses an improved Bayesian framework to automatically learn the parameters of the model [19]. Further, a parking vacancy prediction model named DWT-Bi-LSTM is recently proposed, which combines wavelet transform (WT) and bi-directional long-short term memory (Bi-LSTM) [20] to decompose time series at multiple scales and extract bi-directional temporal features, respectively. Although these methods are able to extract non-linear and high-dimension patterns, the interpretability of the extracted patterns remains challenging. Thus, a data augmentation method that can transform human expertise in parking behaviour into prior knowledge of the prediction method is required.

In the last decade, machine learning methods have attracted much attention to traffic condition prediction due to their excellent ability to model complex relationships between features [21–23]. The forecasting task for parking occupancy is no exception. Thus, in addition to data augmentation, network structure optimization has become another important solution. For example a support vector regression (SVR) method with fruit fly optimization [24], whose parameters are updated using *Drosophila* populations and fruit fly optimization algorithm. In order to improve the method's learning ability and generalizability, deep learning methods have become increasingly popular. A method [24] that utilizes the LSTM to extract temporal feature is a representative example. Furthermore, many researchers try to combine neural networks for accuracy improvement, e.g. a hybrid POP method [25] which stacks gate recurrent unit (GRU) and Long short-term memory (LSTM) together. This method combines the advantages of LSTM and GRU in terms of prediction accuracy and prediction efficiency, respectively. Similarly, a hybrid deep learning method, named dConvLSTM-DCN [26], is proposed to make short-term (within 30 min) and long-term (over 30 min) predictions on the vacant parking space availability by zone. Specifically, it can effectively capture the temporal correlations of different time scales and the spatial correlations among different parking lots by the two parallel ConvLSTM components. An illustration in other field is RF-CWRNN [27], a method for multi-lane speed forecasting combining clockwork recurrent neural network and random forest. Although these RNN-based models have contributed to outstanding improvement in accuracy, a flexible weight assignment of the dependency in time series data remains challenges, especially the long-term ones. To achieve the optimization, some recent studies have recently proposed deep learning

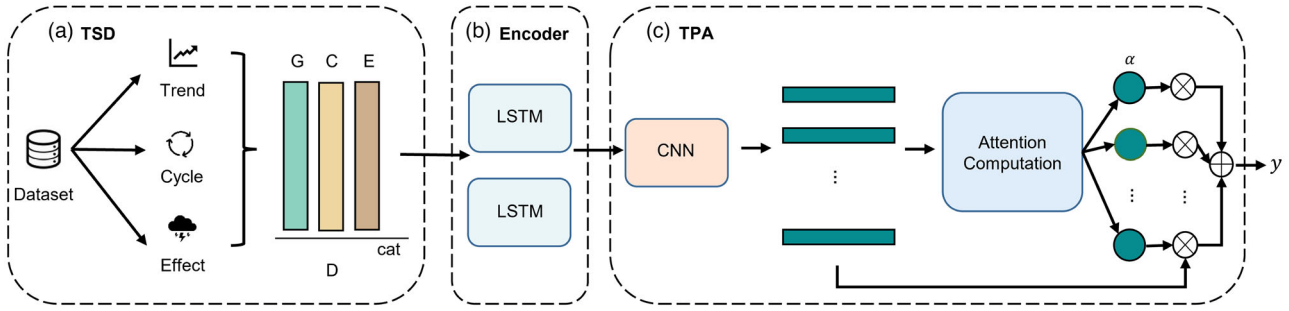


FIGURE 1 Overview of the proposed model structure.

models with various attention mechanisms [28] to achieve better feature representation. For example [29] designed an attention-based model for online taxi-hailing demand forecasting, which helped to better assign weights to spatial and temporal feature maps. However, limited studies constructed an attention-based model for POP. On the other hand, even though the typical attention mechanism reviews the information at each previous time step and selects relevant information to help generate the outputs; however, it fails to capture temporal patterns across multiple time series. Even though the parking utilization, in general, remains a stable periodicity, e.g. peaks and off-peaks, day and night, weekdays and weekends, they may also vary along with special events and holidays. Moreover, even in the same context, the parking type of parking facilities may also affect the occupancy, e.g. shopping centers may have a higher parking ratio on Valentine's Day than hospitals [30].

In summary, although existing deep neural networks outperform statistical methods and traditional machine learning methods in accuracy, they suffer from misinterpretations and fail to capture temporal patterns across multiple time series. To tackle these problems, we propose an integrated parking occupancy prediction method with TSD, LSTM, and TPA. Compared to previous studies, the proposed model is more interpretable by modelling the time series of parking occupancy into three features, namely trend, cycle, and effect. Moreover, a corresponding network with the advanced temporal attention mechanism is designed, which can assign appropriate attentions on the above-mentioned features both internally and cross-wise to achieve state-of-the-art prediction performance in accuracy.

3 | METHODOLOGY

As shown in Figure 1, our method consists of three modules, namely a) data augmentation based on TSD, which decomposes data to three time-series features, trend, cycle and effect; b) an encoder module with two-layer LSTM, which is used to extract high-dimension temporal hidden patterns; c) an attention module with TPA, which can capture temporal patterns across multiple time steps and assign attention to each feature sequence. Details of our method are discussed in the following subsections.

3.1 | The TSD module

Parking occupancy shows three main features of change. First, as a time series, it has a growth trend, such as upward and downward trends during rush hours. Second, the changes in parking occupancy are distinctly cyclical due to commuting behaviours during the day and night or workdays and off days. At the same time, they also show a clear tendency to be affected by factors such as holidays, special events, and traffic accidents. Therefore, we adopt the conception of time-series decomposition to decompose three features from parking occupancy, namely the trend term G , the cycle term C , and the effect term E , using the look-back mechanism, Fourier series fitting, and an indicator function, respectively. The trend, cycle, and effect terms represent the current, cyclical and special changes in the parking occupancy, respectively. These three features will be concatenated as the inputs to the prediction, as presented in Formula (1). Noted that the bold symbol indicates that the value is a vector.

$$y = F(\mathbf{G}, \mathbf{C}, \mathbf{E}) \quad (1)$$

where F is the function from features to predicted value; y is the predicted value.

3.1.1 | Trend

First, the trend term is used to model the current growth trend of parking occupancy. Since future trend depends on previous moments, we utilize the look-back mechanism [31] to define the trend term as Formula (2).

$$\mathbf{G} = [x_{t-w}, x_{t-w+1}, \dots, x_t] \quad (2)$$

where x_t represents the occupancy at time t , and w is the look-back window size.

3.1.2 | Cycle

Second, we use Fourier series to fit the parking occupancy to extract the cyclical feature. Assuming the cycle length of a func-

tion $f(\mathcal{Z})$ is ω , the standard Fourier series expansion of $f(\mathcal{Z})$ can be written as Equation (3).

$$f(\mathcal{Z}) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{2\pi n \mathcal{Z}}{\omega} + b_n \sin \frac{2\pi n \mathcal{Z}}{\omega} \right) \quad (3)$$

where $\mathcal{Z}$ is the independent variable; n is the number of triangle signals; a_0 is a constant coefficient; a_n and b_n are the coefficients of sine and cosine, respectively.

Theoretically, the Fourier series has an infinite number of sine and cosine signals. However, in practice, we only need to take enough sine and cosine signals to fit the target function within a given error range. So we specify M as the number of signals. Then, the cycle value c_t at time t can be written as Equation (4).

$$c_t = \frac{a_0}{2} + \sum_{n=1}^M \left(a_n \cos \frac{2\pi nt}{T} + b_n \sin \frac{2\pi nt}{T} \right) \quad (4)$$

where T is the cycle length, which is a hyper-parameter; t is the time step.

For simplicity, we split out the coefficients and triangle signals in Equation (4) to two vectors, i.e. $\mathcal{A}$ and $\mathcal{B}$. So that we can rewrite it to a linear regression task, as Equation (5).

$$\begin{aligned} \mathcal{A} &= [a_1, b_1, \dots, a_n, b_n]^T \\ \mathcal{B} &= \left[\cos \frac{2\pi n}{T}, \sin \frac{2\pi n}{T}, \dots, \cos \frac{2\pi mn}{T}, \sin \frac{2\pi mn}{T} \right] \\ c_t &= \frac{a_0}{2} + \mathcal{B}\mathcal{A} \end{aligned} \quad (5)$$

We solve this multiple linear regression task by batch gradient descent (BGD). Then, the cycle term $\mathcal{C}$ can be defined as Equation (6).

$$\mathcal{C} = [c_{t+1}, c_{t+2}, \dots, c_{t+w+1}] \quad (6)$$

where c_t represents the value of Fourier series expansion at time t , and w is the cycle window size that is equal to the look-back window size in Equation (2).

3.1.3 | Effect

Changes in parking occupancy are influenced by external factors, such as special events, holidays, weather etc. In order to automatically model the effect caused by the above factors, we build an indicator function calculated by trend and cycle to describe effect $\mathbf{E}$, whose element e_t at time t is defined as Equation (7). It shows the degree the trend deviates from cycle.

$$e_t = \frac{(x_t - c_t)}{x_t} \quad (7)$$

Further, to constitute the effect vector $\mathbf{E}$, the calculation can be written as Equation (8).

$$\mathbf{E} = [e_{t-w}, e_{t-w+1}, \dots, e_t] \quad (8)$$

where e_t represents the value of E at time t , w is the window size.

Finally, the input feature term $\mathbf{D}$ is concatenated by $\mathbf{G}$, $\mathbf{C}$, and $\mathbf{E}$, which can be expressed as Equation (9).

$$\mathbf{D} = \begin{bmatrix} \mathbf{G} \\ \mathbf{C} \\ \mathbf{E} \end{bmatrix} = \begin{bmatrix} x_{t-w}, x_{t-w-1}, \dots, x_t \\ c_{t+1}, c_{t+2}, \dots, c_{t+w+1} \\ e_{t-w}, e_{t-w-1}, \dots, e_t \end{bmatrix} \quad (9)$$

In summary, by using the backtracking mechanism of RNNs to define the trend vector, the trend term can model near-future changes, e.g. upward, downward, turning, and levelling; by fitting a Fourier series consisting of a set of triangular signals, the cycle term can simulate the multiple periodic patterns in occupancy changes, including peaks and off-peaks, day and night, weekdays and weekends; by telling how much the current occupancy deviates from the “cycle”, the effect term can express the occupancy changes caused by arranged or casual events, e.g. holidays, extreme weather, and special events.

3.2 | Encoder module

To extract high-dimensional temporal patterns, we use a two-layer LSTM to build the encoder module. LSTM is an optimized recurrent neural network (RNN), which can cope with the correlation within time series in both short and long term [32].

LSTM generally defines a recurrent function F_L and calculates b_t for current time t , as Equation (10).

$$b_t, s_t = F_L(b_{t-1}, s_{t-1}, f_t) \quad (10)$$

where $b_t \in R^m$ is the hidden layer state of the LSTM at time t , m is the dimension of variables; $s_t \in R^m$ is the final state of memory cell at time t ; d_t is the value of feature sequence at time t .

The calculation is defined by Formula (11):

$$\begin{aligned} i_t &= \text{sigmoid}(W_{d_i} d_t + W_{b_i} b_{t-1}) \\ f_t &= \text{sigmoid}(W_{d_f} d_t + W_{b_f} b_{t-1}) \\ o_t &= \text{sigmoid}(W_{d_o} d_t + W_{b_o} b_{t-1}) \\ s_t &= f_t \odot s_{t-1} + i_t \odot \tanh(W_{d_g} d_t + W_{b_g} b_{t-1}) \\ b_t &= o_t \odot \tanh(s_t) \end{aligned} \quad (11)$$

where t is the time step, i_t is input gate, f_t is forget gate, o_t is output gate. i_t , f_t , and $o_t \in R^m$. W_{d_i} , W_{d_f} , W_{d_o} , W_{d_g} , W_{b_i} , W_{b_f} , W_{b_o} , and W_{b_g} are coefficient matrix. W_{d_i} , W_{d_f} , W_{d_o} , and

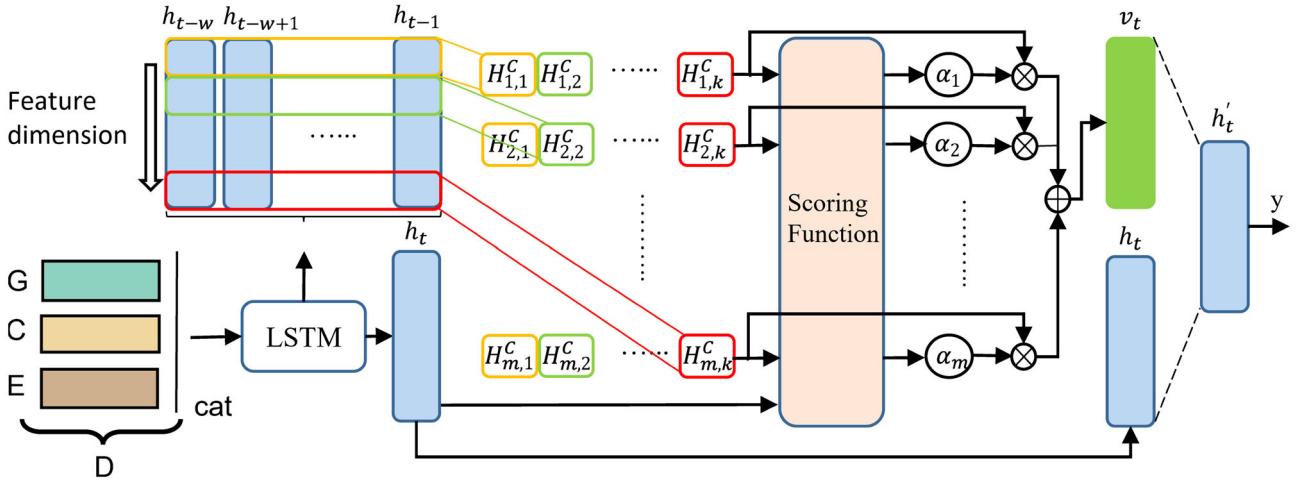


FIGURE 2 Network structure of the TPA module.

$W_{d_g} \in R^{m \times n}$, m, n are the dimension of variable. $W_{b_i}, W_{b_f}, W_{b_o}$, and $W_{b_g} \in R^{m \times m}$, and $\odot$ denotes elements-wise multiplication.

3.3 | TPA module

The features that have a greater impact on the prediction results are assigned greater weights, thus improving the prediction accuracy of parking occupancy.

The typical attention mechanism selects information from previous time steps relevant to the current time step. If there are multiple variables in each time step, it fails to select the variables relevant to the current time step in terms of prediction utility. Moreover, since the typical attention mechanism averages the information across multiple time steps, it fails to detect temporal patterns useful for prediction.

In the TPA module, the attention mechanism basically attends to the row vectors of the state matrix of LSTM and the attention weights on rows select those variables that are helpful for prediction. According to the state matrix of the hidden layer of LSTM, the TPA first detects temporal patterns using CNN, then calculates the context vector v_t with attention mechanism and makes prediction with the context vector v_t at last.

As is shown in Figure 2, the state matrix H consists of previous hidden layer states, which is presented as Equation (12).

$$H = [h_{t-w}, h_{t-w+1}, \dots, h_{t-1}] \quad (12)$$

where t is the current moment, w is the window size.

3.3.1 | Temporal pattern detection using CNN

In the TPA module, CNN is used to enhance the learning ability of the method by applying CNN filters on the row vectors of state matrix H . The state matrix H applies one-dimensional

CNN filters on its row vectors to detect the temporal pattern in the feature series. As is shown in Figure 2, there are k one-dimensional CNN filters $C_i \in R^{1 \times w}$ with length w . The rectangles in different colours represent different filters. Each filter convolves over m features of hidden layer states and produces a matrix H^C with m rows and k columns. $H_{i,j}^C$ represents the convolutional value of the i th row vector and the j th filter. The operation is given as Equation (13).

$$H_{i,j}^C = \sum_{l=1}^w H_{i,(t-w-1+l)} \times C_{j,l} \quad (13)$$

3.3.2 | Attention mechanism

In the part of the attention mechanism, the TPA calculates the context vector v_t to make the prediction. First, the scoring function calculates a weight for each row of H^C by comparing it with the current hidden layer state h_t . Then, the context vector v_t is calculated by the weighted sum of the row vectors of H^C . Then, the final prediction value is calculated with the context vector v_t . The scoring function F_f is defined as Equation (14).

$$F_f(H^C, h_t) = H_i^C W_a h_t \quad (14)$$

where H_i^C is the i th row of H^C , W_a is coefficient matrix and $W_a \in R^{k \times m}$. The attention weight of the i th row vector of H^C , α_i , is calculated as Equation (15).

$$\alpha_i = \text{sigmoid}(F_f(H^C, h_t)) \quad (15)$$

Then, the row vectors of H^C are weighted by α_i to calculate the context vector $v_t \in R^k$.

$$v_t = \sum_{i=1}^m \alpha_i H_i^C \quad (16)$$

ALGORITHM 1 The proposed approach for POP**Input:** The vector of sampled parking lot occupancy $X = \{x_1, x_2, \dots, x_n\}$ **Output:** The predicted value $\hat{y}_i$

- 1: Utilize the look-back mechanism to define the trend term G as Formula (2)
- 2: **for** epoch $\in [0, \text{max epochs1}]$ **do**
- 3: Train a linear regression task, as Equation (5)
- 4: **end for**
- 5: Calculate the cycle term C as Equation (6)
- 6: Calculate the effect vector E as Equation (7) (8)
- 7: **for** epoch $\in [0, \text{max epochs2}]$ **do**
- 8: Train a TSD-LSTM-TPA model
- 9: **end for**
- 10: **for** parking lot $\in$ parking lots **do**
- 11: predict the occupancy of a parking lot with the TSD-LSTM-TPA model
- 12: **end for**
- 13: Output the predicted value $\hat{y}_i$ and calculate the evaluation metric, i.e. RMSE, MAPE, MAE, RAE, R2

Then, we integrate v_t and b_t to predict the target value.

$$b'_t = W_b b_t + W_v (v_t)^T \quad (17)$$

$$y = W_{b'} b'_t \quad (18)$$

where b'_t is integrated vector, y is the predictive value, W_b , W_v and $W_{b'}$ are coefficient matrix, b_t and $b'_t \in R^m$, $W_b \in R^{m \times m}$, $W_v \in R^{m \times k}$, $W_{b'} \in R^{1 \times m}$.

In summary, we employ the conception of time series decomposition to split the occupancy into three interpretable features, and design an effective network with temporal pattern attention mechanism to model the internal and cross-wise patterns of the three features. The computational procedure of TSD-LSTM-TPA is illustrated in pseudocode as shown in Algorithm 1.

4 | EXPERIMENTAL PERFORMANCE

In this section, the proposed method will be tested together with other representative prediction methods based on the same dataset and evaluation metrics. Moreover, the results will be analyzed to demonstrate the improvements achieved.

4.1 | Data description

We evaluate the method performance in a shared dataset with a minimum resolution of 5 min, which is created based on the parking occupancy data of 30 parking lots located in Guangzhou, China, as shown in Figure 3. There are ten parking lots for commercial buildings, three parking lots for hospitals,

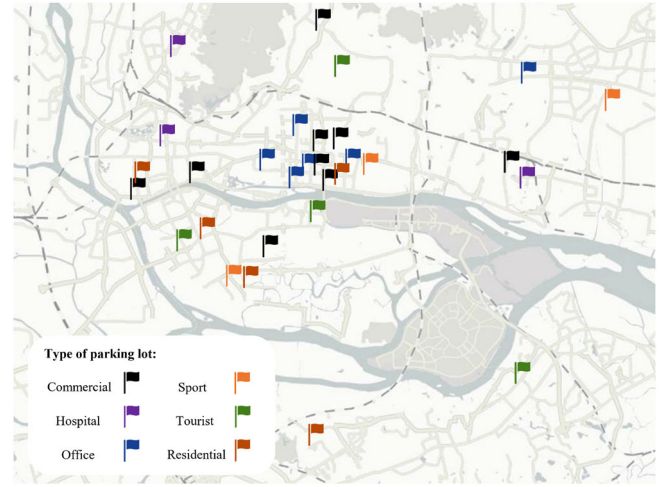


FIGURE 3 Distribution of parking lots in the dataset.

TABLE 1 Statistics of the dataset.

Type	Number	ave.	std.	min.	max.
Commercial	10	0.49	0.19	0.07	1.00
Hospital	3	0.57	0.22	0.07	1.00
Office	5	0.52	0.18	0.09	1.00
Residential	5	0.71	0.20	0.04	1.00
Sport	3	0.29	0.13	0.14	0.85
Tourist	4	0.49	0.27	0.10	1.00
Total	30	0.52	0.23	0.04	1.00

five parking lots for office buildings, three parking lots for sports and recreational facilities, four parking lots for tourist sites, and five parking lots for residential areas. The usage records of these parks are organized as a set of parking occupancies updated every 5 min, which are summarized in Table 1. From the table, we can see that the occupancy of parking lots located in residential areas is the highest, with an average value of 0.71, while the ones in tourist attractions fluctuates the most sharply, with a standard deviation (STD) of 0.27. Moreover, all the studied parking lots have an average occupancy of around 0.52, fluctuating around 0.23, with the exception of the parking lot for sports and recreational facilities, which only has an average occupancy of 0.29. Experiments based on this dataset allow to assess the robustness and generalizability of the comparative models.

4.2 | Evaluation metrics

In order to make a full and fair comparison, five metrics, i.e. mean absolute percentage error (MAPE), mean absolute error (MAE), root mean squared error (RMSE), relative absolute error (RAE), and R-square coefficient (R^2), are used as the indicators of the performance of models. The calculation formulas are presented in Equation (19), where y_i and $\hat{y}_i$ are the observed

and predicted value of sample i , respectively; $\bar{y}$ is the mean value of samples; and N is sample size.

$$\begin{aligned}
 \text{MAPE} &= \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \\
 \text{MAE} &= \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \\
 \text{RMSE} &= \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \\
 \text{RAE} &= \frac{\sum_{i=1}^N |y_i - \hat{y}_i|}{\sum_{i=1}^N |y_i - \bar{y}|} \\
 R^2 &= 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}
 \end{aligned} \tag{19}$$

4.3 | Experimental setup

In order to evaluate the effectiveness of the proposed method, besides a typical statistical regression model, we select a traditional machine learning method and six neural network models as competitive approaches. Specifically, the statistical regression model is least absolute shrinkage and selection operator (Lasso) [33], which retains the advantage of subset shrinkage and is a biased estimator dealing with data with complex covariance. We also evaluate the performance of support vector regression (SVR) [34], which is an application of support vector machine (SVM) for regression problems. Six neural network models are used, namely fully connected neural network (FCNN): a network widely used for function approximation and general regression problems, but it cannot distinguish between heterogeneous features; gated recurrent unit (GRU) [35]: a recurrent neural network with a simplified gate structure and the advantage of fast computation; long short-term memory (LSTM) [32]: a recurrent neural network that is widely used in many time series prediction tasks due to its excellent temporal feature extraction capabilities; bi-directional long short-term memory (BiLSTM) [36]: a combination of forward and backward LSTM, often used for modelling contextual information in natural language processing tasks; temporal convolutional network (TCN) [37]: a time series prediction structure evolved from CNN. It directly exploits the powerful property of convolution to extract features across time steps. MGRU [30]: an recent developed model for parking occupancy prediction that combines empirical mode decomposition and GRU. Whereof, GRU, LSTM, BiLSTM, and MGRU are recurrent neural network (RNN)-based models, while TCN is convolutional neural network (CNN)-based model. To demonstrate the contribution of TSD and TPA modules, LSTM is used as the baseline.

The hyper-parameters and configurations are configured as followings: for each parking lot, a prediction task will be per-

formed separately, in which there are 8640 (30 days $\times$ 288 time steps per day) samples; in Formula (4), the cycle length T is set to be 2016 (12 $\times$ 24 $\times$ 7, which is the number of samples in a cycle of 7 days) when the time interval is 5 min. Moreover, we set the number of triangle signals M to 30, which is found eligible for the data in [38]. Second, the training and test set are divided according to a ratio of 8 to 2 in every parking lot, which means the former 80% data can be used to train the model, and the rest 20% only can be used for evaluation, which is a strict prohibition of information disclosure. Moreover, the test will be conducted to predict the parking occupancy rate in four time intervals, namely 15, 30, 45, and 60 min. Note that their prediction time steps are 3, 6, 9, and 12, respectively, as the temporal resolution of the dataset is 5 min. In all the learning methods except SVR, we use a uniform loss function, mean square error (MSE), as shown in Formula (20). In addition, the input to the proposed model is a three-dimensional tensor with the shape (batch size, sequence length, feature number). Specifically, the batch size is set to 512, which is proper for the dataset; the sequence length is set to 12, representing 1-h historical data; feature number is 3, namely trend, cycle, and effect. For the other models being compared, their batch size and sequence length are the same as the proposed model, while their feature number is 1, i.e. the undecomposed occupancy data. The hyper-parameters in terms of network structure were configured to build the simplest structure of the proposed model, and then the same hyper-parameters will be deployed to other comparative models accordingly, ensuring that each model has the same number of learnable parameters and input information. Besides, the same model-agnostic configurations are deployed to each test, such as learning rate, optimizer, and epoch number, as shown in Table 2.

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \tag{20}$$

Finally, all the experiments are conducted on a Windows workstation with an NVIDIA Quadro RTX 4000 GPU, an Intel(R) Core(TM) i9-10900K CPU, and 64G RAM. To exclude the chance of random seeds, all experiments were performed with 10 replications and averaged. It is worth noting that for reproducibility, the dataset and code used by this paper are shared in Github, and downloadable from the link ¹.

4.4 | Experiment results and discussion

The performance of evaluated methods is analyzed in three parts, namely 1) the competitive experiment to illustrate how well the model is to predict the future; 2) the ablation experiment to verify the 'TSD and TPA modules' effectiveness for accuracy improvement; and 3) the feature importance analysis to explain the influences of all the input features.

¹ <https://github.com/kuanghx3/TSD-LSTM-TPA/tree/main>

TABLE 2 List of running configurations.

Model	Param	Value	Comment
TSD-LSTM-TPA (proposed)	Sequence length	12	—
	Input feature	3	Consisting of trend (12), cycle (12), and effect (12)
	Output feature	1	—
	LSTM layers	2	—
	Learning rate	(0.02, 0.01)	(Fourier fitting, our network)
	Max epochs	(1000, 3000)	(Fourier fitting, our network)
	Optimizer	(BGD, Adam)	(Fourier fitting, our network)
	Loss function	MSE	All
RNNs	Structure	(1, 3, 1, 2)	(Input, hidden, output, layers)
TCN	Structure	(1, [12, 1])	(Input, num_channels) for TCN
FCNN	Structure	(1, 1, 3)	(Input, output, layers) for FCNN
Lasso	Alpha	0.000000001	It is implemented on Scikit-learn (sklearn)
SVR	Kernel, epsilon	RBF, 0.001	It is implemented on Scikit-learn (sklearn)

TABLE 3 The list of evaluation metrics of compared models.

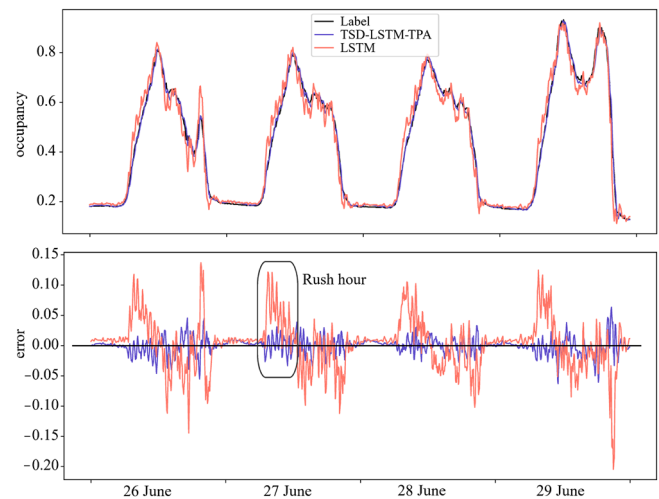
Metric (10^{-2})	RMSE	MAPE	RAE	MAE	R2
TSD-LSTM-TPA	2.11	3.05	16.87	1.47	94.68
MGRU	2.23	3.20	17.68	1.56	94.43
LSTM	2.38	3.30	18.36	1.62	93.43
GRU	2.36	3.27	18.10	1.61	93.69
BiLSTM	2.42	3.39	18.51	1.65	93.43
FCNN	2.48	3.49	18.78	1.70	93.30
Lasso	2.42	3.39	17.49	1.65	94.10
SVR	2.45	3.20	17.79	1.59	93.63
TCN	2.64	3.68	19.11	1.80	93.09

The metrics are averaged based on the results of 120 tasks (30 parking lots $\times$ 4 prediction time intervals, i.e. 15, 30, 45, and 60 min).

Bold indicates the best results in each test group, same below.

4.4.1 | Comparison experiment

As shown in Table 3, the evaluation metrics of compared models are summarized. We can see that the proposed method, TSD-LSTM-TPA, reduces the prediction errors significantly with the highest scores in all the five evaluation metrics, namely RMSE 0.0211, MAE 0.0147, MAPE 3.05%, RAE 16.87%, and R2 94.68%. Compared to other methods, the proposed method, on average, improves by about 12.83% in RMSE, 10.64% in MAE, 9.15% in MAPE, 7.36% in RAE and achieves the best fit with less than 5.5% residuals in R2. Compared to the baseline, i.e. LSTM, TSD-LSTM-TPA improves accuracy by 11.53% in RMSE, 9.31% in MAE, 7.58% in MAPE, and 8.12% in RAE, respectively. Compared to the state-of-the-art model, i.e. MGRU, TSD-LSTM-TPA reduces the prediction error by an average of 5.41% across the five metrics. It illustrates that the TSD, LSTM and TPA modules can work jointly and smoothly to achieve the best performance.

**FIGURE 4** Lineplots of the 45-min prediction results and errors for a parking lot located in commercial areas.

As an visual illustration, the 45-min prediction results of a commercial building parking lot is plotted, as shown in Figure 4, considering the average travel time in Guangzhou is close 40 min [39]. It can be seen that the predicted curve of TSD-LSTM-TPA is more consistent with the true value curve than that of the baseline, i.e. LSTM. More clearly, the error plot shows that the proposed model has more stable predictions during periods of occupancy fluctuations, such as during the morning rush hours. The same conclusion can be found in the forecasts on other parking lots, which are the ones for tourist sites, hospital, office buildings, and sports and recreational facilities, respectively, as shown in Figure 5. Even though peak times vary across parking lots, the proposed model provides an accurate forecast on their occupancy fluctuations during the period, which is superior to the typical time-series prediction method LSTM. It can

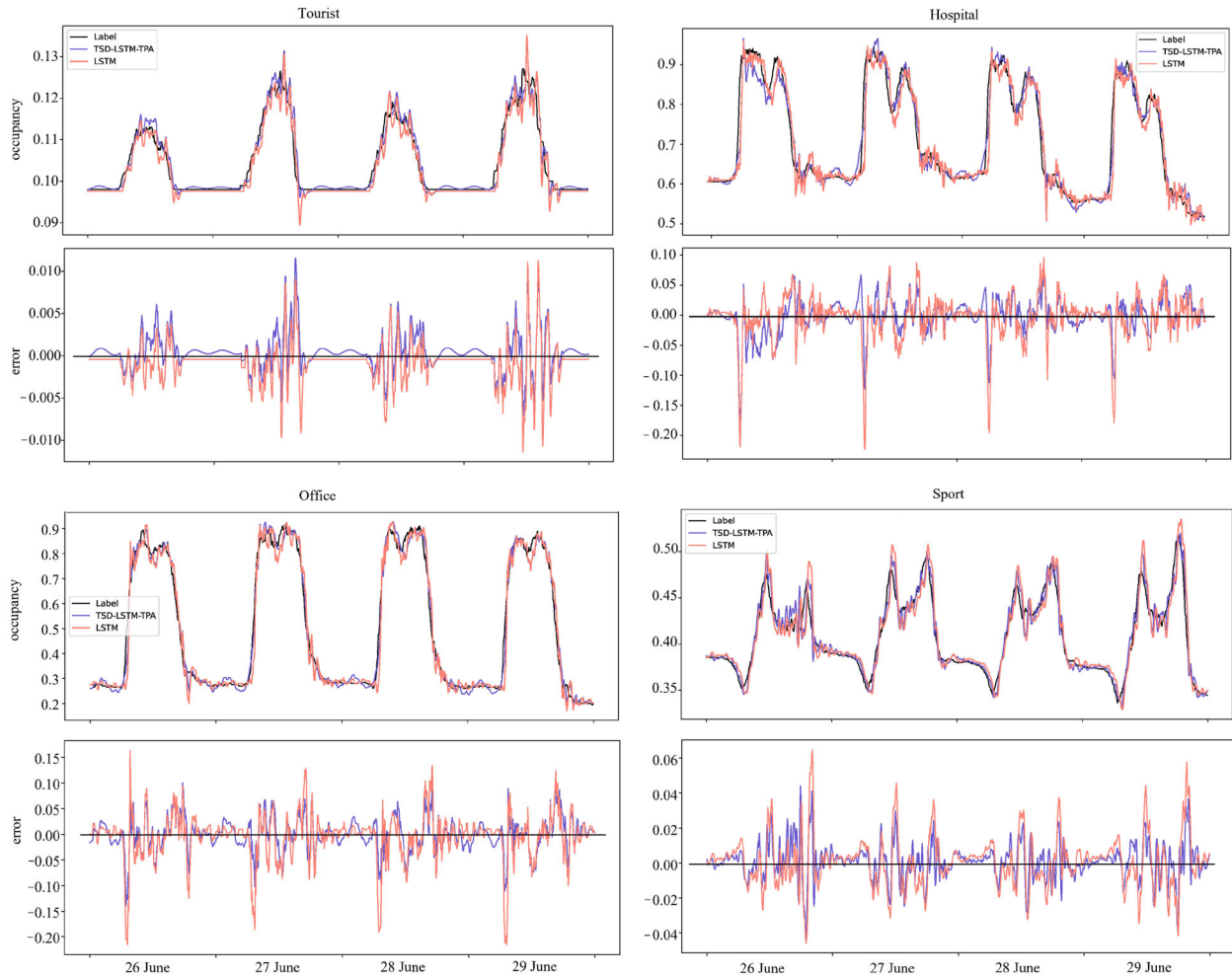


FIGURE 5 Lineplots of the 45-min prediction results and errors for four parking lots located in different areas.

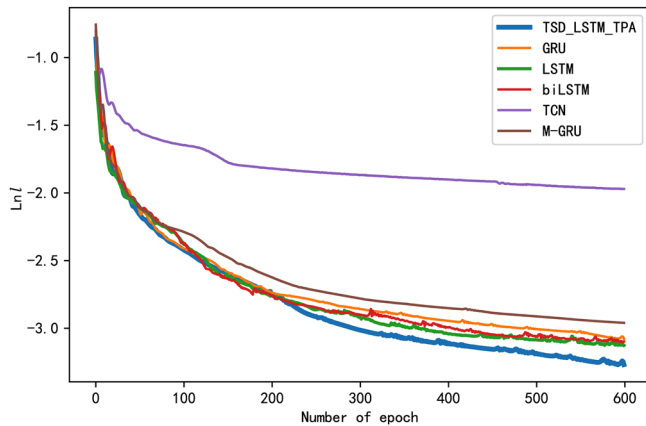


FIGURE 6 The loss curves of compared models in former 600 epochs, where l represents the mean square error.

be empirically concluded that the two optimization (i.e. TSD and TPA) are effective in POP.

The loss decline profiles of compared models are illustrated in Figure 6. From which, it can be seen that the RNN-based

models, namely GRU, LSTM, BiLSTM, and MGRU, are significantly ahead of the CNN-based model, i.e. TCN, in terms of loss decline, demonstrating their superiority in learning time-series knowledge. More importantly, the proposed model has a better prospect for loss decline than the RNN models, especially after 200 iterations, which somehow indicates its excellent learning ability.

As shown in Table 4, we can estimate the generalizability of the models by comparing their ability to tackle different intervals of prediction. First, we found that the statistical method (i.e. Lasso) not only fitted the data quickly but also performed very well in 15 min. It suggests that in an extreme short-term prediction task, the statistical method, whose key idea is catching the trend, is superior to machine learning methods. However, with the increasing prediction intervals, machine learning methods, especially neural networks, can take advantage of their strengths in extracting the heterogeneous, high-dimension, and non-linear features. In this context, the result shows that our model outperforms other neural networks in 30, 45, and 60-min prediction, because we focus more attention on the long-term pattern (i.e. the cycle term). Moreover, thanks to the TPA module, the proposed model can get a set of reasonable weights

TABLE 4 The comparison of models' prediction ability in four intervals .

Metrics (10^{-2})	15-min					30-min				
Model	RMSE	MAPE	RAE	MAE	R2	RMSE	MAPE	RAE	MAE	R2
TSD-LSTM-TPA	1.1944	1.6631	10.3076	0.8246	97.7477	1.8345	2.6009	14.9694	1.2760	95.8845
MGRU	1.3240	1.8593	11.1683	0.9060	97.5799	1.9221	2.7091	15.5921	1.3269	95.7785
LSTM	1.2692	1.7444	10.9353	0.8661	97.1574	2.0236	2.7726	15.8060	1.3810	95.4212
GRU	1.2559	1.7223	10.6033	0.8630	97.6601	1.9826	2.7418	15.5430	1.3553	95.6157
BiLSTM	1.3045	1.8024	10.7294	0.8924	97.7232	2.0579	2.8840	16.2844	1.4090	95.0378
FCNN	1.2769	1.7237	10.5506	0.8667	97.6985	2.0805	2.8689	16.1527	1.4240	95.3271
Lasso	1.2047	1.5897	8.8197	0.8080	98.4956	2.0026	2.7288	14.5780	1.3552	96.1939
SVR	1.2264	1.6658	9.8984	0.8279	98.1614	2.0246	2.6589	15.2104	1.3296	95.7221
TCN	1.3986	1.8520	10.1422	0.9335	98.0374	2.0955	2.7984	17.5068	1.4023	94.5836
Metrics (10^{-2})	45-min					60-min				
Model	RMSE	MAPE	RAE	MAE	R2	RMSE	MAPE	RAE	MAE	R2
TSD-LSTM-TPA	2.4197	3.5267	19.0619	1.6883	93.7879	2.9774	4.4177	23.1514	2.0910	91.2894
MGRU	2.5630	3.6755	20.1509	1.7982	93.4213	3.0974	4.5479	23.8056	2.1898	90.9262
LSTM	2.7402	3.7879	20.7558	1.8610	92.5232	3.4913	4.9041	25.9546	2.3758	88.6139
GRU	2.7223	3.7553	20.8047	1.8579	92.3732	3.4792	4.8794	25.4412	2.3723	89.1226
BiLSTM	2.7532	3.8511	20.5879	1.8752	92.5839	3.5588	5.0343	26.4413	2.4269	88.3756
FCNN	2.8621	4.0341	21.2498	1.9593	92.2799	3.6961	5.3317	27.1553	2.5437	87.9033
Lasso	2.8155	3.9634	20.3514	1.9257	92.9816	3.6597	5.2883	26.2255	2.5192	88.7412
SVR	2.8442	3.7122	20.7701	1.8438	92.2810	3.6866	4.7566	25.2933	2.3640	88.3610
TCN	3.0143	4.5726	22.1246	2.1045	91.8117	4.0578	5.5019	26.6555	2.7718	87.9119

to the three feature sequences (i.e. the trend, cycle, and effect term).

Similarly, we test the models' ability to tackle different types of parking lots as shown in Table 5. The results suggest that the proposed model performs best in the parking lots located in commercial areas, hospitals, office buildings and residential areas, which have significant cyclical patterns. In contrast, the proposed model has no advantage in predicting occupancy for the other two types of parking lots, i.e. sport and tourist sites, where there are significant off-cycle occupancy fluctuations in peak season. Considering that the cycle length in this study is set to 1 week, it is reasonable the proposed model fails to capture the monthly or annual patterns in these two type of parking lots.

4.4.2 | Ablation experiment

In order to verify the effectiveness of the modules (i.e. TSD, LSTM, and TPA) in our model, we conduct an ablation experiment with all the 30 parking lots in all the four prediction intervals, as shown in Table 6. First, TSD-LSTM-TPA is better than t&c-LSTM-TPA (trend and cycle only) and t&e-LSTM-TPA (trend and effect only), which demonstrates the effectiveness of cycle and effect terms. Second, the proposed model performs better than LSTM-TPA, which demonstrates TSD module plays an important role in improving the predic-

tion accuracy. The prediction accuracy of LSTM-TPA decreases 13.27% in RMSE, 9.84% in MAPE, 11.14% in RAE, 10.88% in MAE, 2.06% in R2. Third, TSD-LSTM-TPA is better than TSD-FCNN-TPA, which demonstrates the important role of LSTM in extracting temporal features. Fourth, the experiment results of TSD-LSTM-TPA are just slightly better than that of TSD-LSTM. To emphasize contribution of the TPA module, we conducted a noise attack experiment, where a standard normal distributed noise is added to the current occupancy. Figure 7 shows the average RMSE of the four prediction intervals. We can see that in 80% (24 out of 30) of the parking lots, the proposed model has a lower RMSE than the TSD-LSTM under noise interference. Moreover, the average RMSE difference between the models with and without the TPA module is 23.45%. Thus, it can be assumed that the TPA module contributes in prediction by improving the robustness of our model, as it can assign proper attentions to the useful time-series features. Finally, TSD-LSTM-TPA achieves an improvement on other integrated models by 8.87% on average, revealing the integration of the three modules works.

4.4.3 | Feature importance

Understanding why a method makes a certain prediction can be as crucial as the prediction's accuracy in many applications.

TABLE 5 The comparison of models' prediction ability in six types of parking lots.

Metrics (10^{-2})	Commercial (10)					Hospital (3)				
Model	RMSE	MAPE	RAE	MAE	R2	RMSE	MAPE	RAE	MAE	R2
TSD-LSTM-TPA	2.1690	3.0260	15.9939	1.5008	94.5575	2.7550	3.4432	13.8056	1.8989	96.6733
MGRU	2.2552	3.1445	16.5352	1.5577	94.2703	3.0434	3.7988	15.3819	2.1113	96.0891
LSTM	2.5319	3.5117	18.4603	1.7333	92.7950	3.2581	3.7943	15.1165	2.1052	95.3289
GRU	2.5078	3.5063	18.2289	1.7186	92.9751	3.2808	3.8570	15.3717	2.1400	95.1928
BiLSTM	2.5540	3.6046	18.4929	1.7493	92.7851	3.3519	3.9677	15.7740	2.1911	95.0424
FCNN	2.7150	4.0448	19.8972	1.9118	91.9934	3.5260	4.0084	16.2707	2.2599	94.5245
Lasso	2.7054	4.0299	19.4110	1.8951	92.3476	3.3815	3.8649	15.3613	2.1502	94.9277
SVR	2.6170	3.5693	18.1214	1.7409	92.6031	3.5722	3.6354	15.1773	2.1208	94.3205
TCN	2.5500	3.6459	19.9083	1.7805	91.9414	3.9818	4.8510	17.5534	2.5348	93.5040
Metrics (10^{-2})	Office (5)					Sport (3)				
Model	RMSE	MAPE	RAE	MAE	R2	RMSE	MAPE	RAE	MAE	R2
TSD-LSTM-TPA	2.7957	3.9351	20.9733	2.0064	93.4514	1.8166	2.8937	11.7455	1.2803	97.9360
MGRU	2.9289	4.0916	22.4327	2.1049	92.5848	1.9345	2.9874	12.8830	1.3822	97.5565
LSTM	3.1772	4.3284	22.3758	2.2122	92.3930	1.8483	2.7992	12.4312	1.3222	97.8297
GRU	3.1850	4.3892	22.6213	2.2280	92.2374	1.7762	2.6480	11.9429	1.2706	97.9860
BiLSTM	3.1741	4.3410	22.2145	2.2107	92.5752	1.9498	3.0245	12.9306	1.3993	97.6414
FCNN	3.2656	4.4655	22.5103	2.2630	92.3499	1.7606	2.5362	11.6496	1.2269	97.9664
Lasso	3.2583	4.4556	21.8752	2.2580	92.7897	1.6317	2.3702	10.7577	1.1421	98.2666
SVR	3.3060	4.1476	21.6485	2.1902	92.6615	1.7027	2.3579	10.8254	1.1550	98.1326
TCN	3.1557	4.1757	21.9142	2.1927	92.8574	1.2563	2.9223	21.5440	0.8943	92.7852
Metrics (10^{-2})	Tourist (4)					Residential (5)				
Model	RMSE	MAPE	RAE	MAE	R2	RMSE	MAPE	RAE	MAE	R2
TSD-LSTM-TPA	1.0683	2.5810	24.5662	0.7258	89.1170	1.7433	2.2717	16.8821	1.1962	95.1097
MGRU	1.1436	2.8836	24.9962	0.8149	90.6438	1.8423	2.2630	16.8282	1.2164	94.7967
LSTM	1.0102	2.3446	24.2236	0.7039	88.4907	2.0454	2.4740	18.5575	1.2992	93.0864
GRU	0.9860	2.2332	22.9558	0.6855	90.1480	2.0289	2.4302	18.2125	1.3020	93.4741
BiLSTM	1.0302	2.3508	23.9097	0.7303	90.0684	2.0633	2.4898	18.9042	1.3046	92.1593
FCNN	1.0132	2.3430	22.8063	0.7060	90.8131	2.1170	2.5442	19.0775	1.3719	92.8844
Lasso	0.9411	2.0478	16.8718	0.6526	95.1373	2.0366	2.4024	17.7090	1.3003	93.5349
SVR	0.9836	2.1274	21.5548	0.6621	91.7781	2.1205	2.6106	20.0030	1.3141	92.6611
TCN	2.0357	2.1210	17.1517	1.2656	93.3526	1.9224	3.0703	11.4868	1.3338	98.0287

TABLE 6 The list of metrics in ablation experiment.

Metric (10^{-2})	RMSE	MAPE	RAE	MAE	R2
TSD-LSTM-TPA	2.11	3.05	16.87	1.47	94.68
TSD-LSTM	2.11	3.06	16.88	1.48	94.54
t&c-LSTM-TPA	2.20	3.14	17.48	1.54	94.30
t&e-LSTM-TPA	2.26	3.29	17.81	1.56	93.57
LSTM-TPA	2.39	3.35	18.75	1.63	92.73
TSD-FCNN-TPA	2.69	3.95	19.25	1.74	91.50

The metrics are averaged based on the results of 120 tasks (30 parking lots $\times$ 4 forecasting time intervals, i.e. 15, 30, 45, and 60 min).

To address this problem, we adopt a novel framework for interpreting predictions, SHAP [15], which is an explanatory metric of an additive model based on cooperative game theory, where all features are considered as “contributors”. In this section, 36 (3 time-series features $\times$ 12 time steps per feature) SHAP values are calculated to quantify the importance for each input feature in the 45-min prediction, as shown in Figure 8. According to the definition [15], the greater the absolute SHAP value of a specific feature, the more significant its contribution in the prediction. Moreover, the positive SHAP value indicates that the corresponding feature increases the predicted value and conversely, making it smaller. We can see the trend and cycle terms

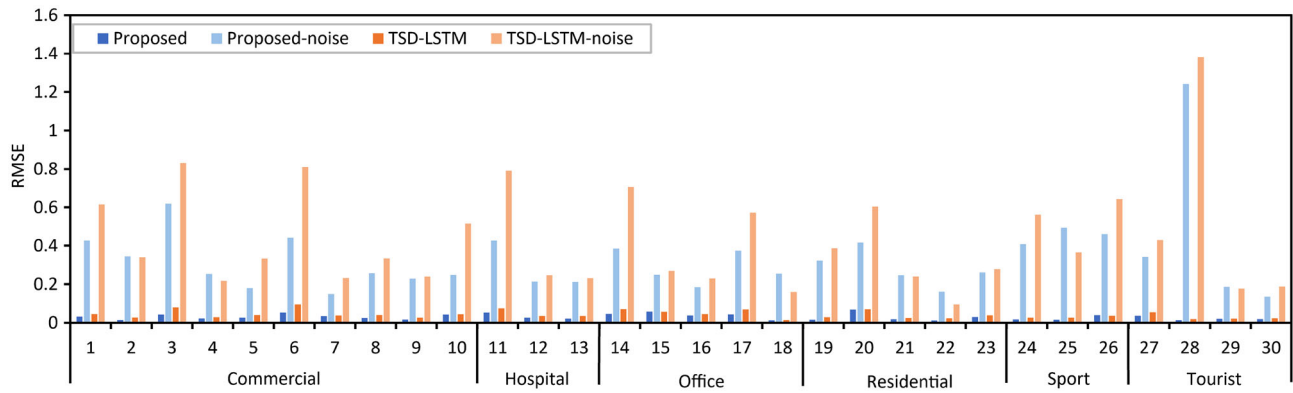


FIGURE 7 Results of the noise attack experiment.

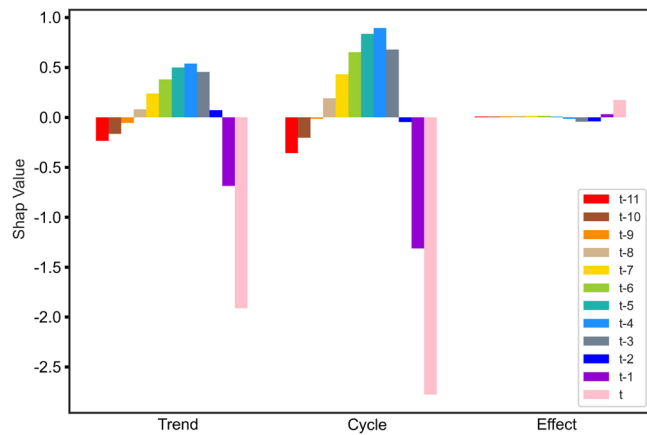


FIGURE 8 The SHAP values of input features in 45-min prediction.

play major roles in prediction, while the impact of effect terms is small but serves a purpose. It yields a similar conclusion to the ablation experiment. From the aspect of time steps, the trend and cycle terms have a similar pattern. We can see the predicted value is most influenced by the two closest moments of the trend and cycle to the present which has a “minus” impact, making the predicted value smaller. Moreover, for the trend and cycle terms, the SHAP values of the features from $(t - 3)$ to $(t - 8)$ are positive, while from $(t - 9)$ to $(t - 11)$ are negative. It indicates that in this study, the past 15 to 40 min of occupancy data would lead the model to overestimate the future occupancy, while the most recent 15 min of data would lead the model to give a conservative estimate. Considering all these information together, our model can get a reasonable prediction result.

In summary, the combination of TSD in feature engineering, LSTM in network design, and TPA in multi-time-series feature attention is efficient and effective. As shown by the evaluation results, the proposed approach can 1) reduce the prediction errors significantly with the highest scores in all the five evaluation metrics, i.e. RMSE 0.0211, MAE 0.0147, MAPE 3.05%, RAE 16.87%, and R2 94.68%; 2) adapt in different forecasting intervals and different types of parking lots, which demonstrates its generalizability. In addition, we estimate the

influence of all the input features, which is helpful for us to explain the prediction.

5 | CONCLUSION

In order to support a near real-time parking occupancy prediction, we propose an integrated approach, which consists of three modules, i.e. TSD, LSTM and TPA. Whereof, the TSD framework is used to model the time series information of various parking lots by extracting the features of trend, cycle, and effect. On this basis, a network combining LSTM and TPA is designed to process the decomposed features, in which LSTM is adopted to extract temporal patterns, and TPA gives a set of reasonable weights for them.

As shown by the evaluation results, the proposed model outperforms other models in two aspects, namely 1) TSD-LSTM-TPA improves accuracy over the baseline model LSTM by 11.53% in RMSE, 9.31% in MAE, 7.58% in MAPE, and 8.12% in RAE, respectively; 2) The generality of the model is enhanced to handle a wide range of prediction time intervals and parking lot types. This is reflected in the consistent and stable performance achieved in each set of tests. Besides, we conduct the ablation experiment and utilize SHAP value to prove the prediction of our model is in line with common sense.

Future work can be conducted in three directions: 1) to introduce multi-source data and evaluate their effectiveness in prediction; 2) to tackle the regional parking occupancy prediction, which predicts multi-parking occupancy jointly; 3) to apply our method in other traffic condition prediction scenarios.

AUTHOR CONTRIBUTIONS

Wei Ye: Conceptualization; data curation; formal analysis; investigation; methodology; software; validation; visualization; writing - original draft. Haoxuan Kuang: Data curation; software; validation; visualization; writing - review and editing. Jun Li: Conceptualization; funding acquisition; project administration; resources; supervision; validation; writing - review and editing. Xinjun Lai: Validation; writing - review and editing. Haohao Qu: Validation; visualization; writing - review and editing.

ACKNOWLEDGEMENTS

The study was supported by Guangdong Key Areas R&D Program (No.2019B090913001), which is gratefully acknowledged.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Github at <https://github.com/kuanghx3/TSD-LSTM-TPA/tree/main>

ORCID

Wei Ye  <https://orcid.org/0000-0002-8655-2011>

REFERENCES

- Giuffrè, T., Siniscalchi, S.M., Tesoriere, G.: A novel architecture of parking management for smart cities. *Proc. Soc. Behav. Sci.* 53, 16–28 (2012)
- Shoup, D.C.: Cruising for parking. *Transp. Policy* 13(6), 479–486 (2006) parking
- Bock, F., Di Martino, S., Origlia, A.: Smart parking: Using a crowd of taxis to sense on-street parking space availability. *IEEE Trans. Intell. Transp. Syst.* 21(2), 496–508 (2020)
- Qin, H., Pang, Q., Yu, B., Wang, Z.: Analysis on cruising process for on-street parking using an spectral clustering method. *IET Intel. Transport Syst.* 14(14), 2113–2121 (2020)
- Caicedo, F.: Real-time parking information management to reduce search time, vehicle displacement and emissions. *Transp. Res. D: Transp. Environ.* 15(4), 228–234 (2010)
- Ma, R., Chen, S., Zhang, H.M.: Time series relations between parking garage occupancy and traffic speed in macroscopic downtown areas - a data driven study. *Int. J. Intell. Transp. Syst.* 25(4), 423–438 (2021)
- Chen, X., He, Z., Chen, Y., Lu, Y., Wang, J.: Missing traffic data imputation and pattern discovery with a Bayesian augmented tensor factorization model. *Transp. Res. Part C: Emerg. Technol.* 104, 66–77. doi:<https://www.sciencedirect.com/science/article/pii/S0968090X18307757>
- Pozo, R.F., González, A.B.R., Wilby, M.R., Vinagre Díaz, J.J., Matesanz, M.V.: Prediction of on-street parking level of service based on random undersampling decision trees. *IEEE Trans. Intell. Transp. Syst.* 23(7), 8327–8336 (2022)
- Sun, Y., Peng, L., Li, H., Sun, M.: Exploration on spatiotemporal data repairing of parking lots based on recurrent GANs. In: 2018 21st International Conference on Intelligent Transportation Systems (ITSC), pp. 467–472. IEEE, Piscataway, NJ (2018)
- Qu, H., Liu, S., Guo, Z., You, L., Li, J.: Improving parking occupancy prediction in poor data conditions through customization and learning to learn. In: *Knowledge Science, Engineering and Management*, pp. 159–172. Springer, Cham (2022)
- Zeng, C., Ma, C., Wang, K., Cui, Z.: Parking occupancy prediction method based on multi factors and stacked GRU-LSTM. *IEEE Access* 10, 47361–47370 (2022)
- Yang, S., Ma, W., Pi, X., Qian, S.: A deep learning approach to real-time parking occupancy prediction in transportation networks incorporating multiple spatio-temporal data sources. *Transp. Res. Part C: Emerg. Technol.* 107, 248–265 (2019). doi:<https://www.sciencedirect.com/science/article/pii/S0968090X18313780>
- Taylor, S.J., Letham, B.: Forecasting at scale. *Am. Stat.* 72(1), 37–45 (2017)
- Shih, S.Y., Sun, F.K., Lee, H.-y.: Temporal pattern attention for multivariate time series forecasting. *Mach. Learn.* 108, 1421–1441 (2019)
- Lundberg, S.M., Lee, S.I.: A unified approach to interpreting model predictions. In: *Proceedings of the 31st Conference on Neural Information Processing Systems (NeurIPS)*, pp. 1–10. Curran Associates, Red Hook, NY (2017)
- Pozo, R.F., González, A.B.R., Wilby, M.R., Díaz, J.J.V., Matesanz, M.V.: Prediction of on-street parking level of service based on random undersampling decision trees. *IEEE Trans. Intell. Transp. Syst.* 23(7), 8327–8336 (2022)
- Fan, G.F., Peng, L.L., Hong, W.C.: Short-term load forecasting based on empirical wavelet transform and random forest. *Electr. Eng.* 104, 4433–4449 (2022)
- Agrawal, R., Kothari, P.: Copasample: A heuristics based covariance preserving data augmentation. In: *Machine Learning, Optimization, and Data Science*, pp. 308–320. Springer, Cham (2019)
- Chen, X., He, Z., Chen, Y., Lu, Y., Wang, J.: Missing traffic data imputation and pattern discovery with a Bayesian augmented tensor factorization model. *Transp. Res. Part C: Emerg. Technol.* 104, 66–77 (2019). doi:<https://www.sciencedirect.com/science/article/pii/S0968090X18307757>
- Zeng, C., Ma, C.X., Wang, K., Cui, Z.H.: Predicting vacant parking space availability: a DWT-Bi-LSTM model. *Phys. A: Stat. Mech. Appl.* 599, 127498 (2022)
- Hsueh Chan, L.E., Chen Hao, L.: A parking occupancy prediction approach based on spatial and temporal analysis. In: *Intelligent Information and Database Systems*, pp. 500–509. Springer, Cham (2018)
- Zhao, Z., Zhang, Y., Zhang, Y.: A comparative study of parking occupancy prediction methods considering parking type and parking scale. *J. Adv. Transp.* 2020, 5624586 (2020). doi:<https://doi.org/10.1155/2020/5624586>
- Liu, Z., Lyu, C., Huo, J., Wang, S., Chen, J.: Gaussian process regression for transportation system estimation and prediction problems: The deformation and a hat kernel. *IEEE Trans. Transp. Syst.* 23(11), 22331–22342 (2022)
- Fan, J., Hu, Q., Xu, Y., Tang, Z.: Predicting vacant parking space availability: A long short-term memory approach. *IEEE Intell. Transp. Syst. Mag.* 14(2), 129–143 (2022)
- Zeng, C., Ma, C., Wang, K., Cui, Z.: Parking occupancy prediction method based on multi factors and stacked GRU-LSTM. *IEEE Access* 10, 47361–47370 (2022)
- Feng, Y., Tang, Z., Xu, Y., Krishnamoorthy, S., Hu, Q.: Predicting vacant parking space availability zone-wisely: a densely connected ConvLSTM method. In: 2021 IEEE Vehicle Power and Propulsion Conference (VPPC), pp. 1–6. IEEE, Piscataway, NJ (2021)
- Lu, W., Yi, Z., Liu, W., Gu, Y., Rui, Y., Ran, B.: Efficient deep learning based method for multi-lane speed forecasting: a case study in Beijing. *IET Intel. Transport Syst.* 14(14), 2073–2082 (2020)
- Do, L.N.N., Vu, H.L., Vo, B.Q., Liu, Z., Phung, D.: An effective spatial-temporal attention based neural network for traffic flow prediction. *Transp. Res. Part C: Emerg. Technol.* 108, 12–28 (2019). doi:<https://www.sciencedirect.com/science/article/pii/S0968090X19301330>
- Liu, Y., Liu, Z., Lyu, C., Ye, J.: Attention-based deep ensemble net for large-scale online taxi-hailing demand prediction. *IEEE Trans. Intell. Transp. Syst.* 21(11), 4798–4807 (2020)
- Li, J., Qu, H., You, L.: An integrated approach for the near real-time parking occupancy prediction. *IEEE Trans. Intell. Transp. Syst.* 24(4), 3769–3778 (2023)
- Elman, J.L.: Finding structure in time. *Cognit. Sci.* 14(2), 179–211 (1990). doi:https://onlinelibrary.wiley.com/doi/abs/10.1207/s15516709cog1402_1
- Hochreiter, S., Schmidhuber, J.: Long short-term memory. *Neural Comput.* 9(8), 1735–1780 (1997). doi:<https://doi.org/10.1162/neco.1997.9.8.1735>
- Tibshirani, R.: Regression shrinkage and selection via the lasso. *J. R. Stat. Soc. Ser. B: Stat. Methodol.* 58(1), 267–288 (1996). doi:<https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/j.2517-6161.1996.tb02080.x>
- Smola, A.J., Scholkopf, B.: A tutorial on support vector regression. *Stat. Comput.* 14, 199–222 (2004)
- Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., et al.: Learning phrase representations using RNN encoder-decoder for statistical machine translation. In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Association for Computational Linguistics, Kerrville, TX (2014)

36. Schuster, M., Paliwal, K.K.: Bidirectional recurrent neural networks. *IEEE Trans. Signal Process.* 45(11), 2673–2681 (1997)
37. Bai, S., Kolter, J.Z., Koltun, V.: An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv:1803.01271* (2018)
38. Qu, H., Liu, S., Li, J., Zhou, Y., Liu, R.: Adaptation and learning to learn (all): an integrated approach for small-sample parking occupancy prediction. *Mathematics* 10(12), 2039 (2022). doi:<https://www.mdpi.com/2227-7390/10/12/2039>
39. Ministry of Housing and Urban-Rural Development of the People's Republic of China (MOHURD), China Academy of Urban Planning and Design, Baidu Map. 2022 Annual Commuting Monitoring Report for

Major Cities in China (2022-07-29)[2022-11-20]. Available from: <https://huiyan.baidu.com/cms/report/2022tongqin/> (2022-07-29).

How to cite this article: Ye, W., Kuang, H., Li, J., Lai, X., Qu, H.: A parking occupancy prediction method incorporating time series decomposition and temporal pattern attention mechanism. *IET Intell. Transp. Syst.* 18, 58–71 (2024). <https://doi.org/10.1049/itr2.12433>